

National University of Computer & Emerging Sciences
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HOSPITAL MANAGEMENT SYSTEM

Project Proposal
Programming Fundamentals
Section: BSE-1B

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Project Proposal

Introduction

Healthcare services often face issues in managing patient appointments, doctor schedules, and hospital information efficiently. A **Hospital Management System (HMS)** is a computerized solution that simplifies the process by allowing patients to find nearby hospitals, book appointments with doctors, and manage schedules.

Our proposed system will allow users to enter their details, detect their location, suggest nearby hospitals, and guide them in booking an appointment based on available doctors and timings.

Existing System

Currently, most hospitals manage appointments either:

- **Manually** – Patients visit the hospital physically or call for appointment booking. This causes delays and inefficiency.
- **Basic Online Systems** – Some hospitals have individual websites or apps, but they are limited to one hospital and often lack features such as location-based hospital suggestions, conflict-free appointment scheduling, and cross-hospital comparisons.

Problem Statement

- Existing systems are **not integrated**; patients must search different hospitals separately.
- Many systems do not provide **location-based hospital suggestions**.
- Appointment booking often fails due to **overlapping schedules** and lack of real-time updates.
- Patients struggle to **choose the right doctor** according to their medical purpose.

Proposed Solution

Our **Hospital Management System** will provide a **unified platform** where:

1. Users first enter their details (name, age, contact info, Location).
2. The system uses location to suggest nearby hospitals.
3. The patient selects a hospital.
4. The system asks the purpose (e.g., general check-up, cardiology, dermatology).
5. Based on the purpose, relevant doctors are suggested.
6. Available time slots are shown (ensuring no overlapping appointments).
7. The system allocates the appointment and confirms it.

This ensures smooth scheduling, location-based hospital discovery, and efficient doctor-patient matching.

Salient Features

- **User-Friendly Interface** for patients to book appointments easily.
- **Location-Based Hospital Suggestions** using user input.
- **Doctor Recommendation System** based on medical purpose.
- **Conflict-Free Scheduling** ensuring no duplicate or overlapping appointments.
- **Hospital & Doctor Database** with specialization and availability.
- **Appointment History & Management** for both patients and doctors.

Tools & Technologies

- **Programming Language:** C
- **Framework:** Not applicable (C is procedural; we'll develop using standard libraries)
- **Frontend / User Interface:** Console-based (text-based menus and prompts)
- **Operating System:** Windows / Linux (cross-platform as C runs on both)